



# Tomasz Młynik

**Date of birth:** 07/09/1992 | **Place of birth:** Gdańsk, Poland | **Nationality:** Polish | **Phone number:**

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## WORK EXPERIENCE

 **CENTRUM NAUKI EXPERYMENT** – GDYNIA, POLAND

**JUNIOR SPECIALIST IN THE EVENTS AND SCIENTIFIC PROGRAMS DEPARTMENT** – 01/08/2025 – CURRENT

- Collaboration with the scientific community, particularly universities and student organisations, building and maintaining a network of contacts,
- Implementation of scientific and popular science projects and programs in collaboration with the educational, cultural, and scientific communities in Poland and Europe,
- Supervision of the organisation and implementation of scientific events at the CNE outside the center
- Collaboration with other CNE departments on joint projects and ongoing activities,
- Preparation of public procurement contracts and agreements for the unit, including the purchase of materials and equipment necessary for the implementation of activities.
- Providing Data Analysis using Python with Machine Learning (Scikit Learn), improving and optimizing internal administrative processes.

 **TRANSLATION AGENCY** – GDYNIA, POLAND

**ADMINISTRATIVE OFFICE ASSISTANT** – 01/11/2016 – 27/09/2018

- Verify data quality from external clients' documents,
- Provide support to customers
- Create Excel data sheet of clients with pipelines for the translation texts,
- Provide invoices to costumer

 **ROBOTY I SPÓŁKA** – GDAŃSK, POLAND

**LECTURER IN PHYSICS** – 01/03/2015 – 26/07/2016

- Lectures on Robotics using Lego Mindstorm (EV3),
- Basics of programming

## EDUCATION AND TRAINING

30/09/2019 – 30/09/2024 Gdańsk, Poland

**PHD** University of Gdańsk

- Conducted scientific research in fields of Theory of Positive maps mainly the problem of characterisation of positive maps on low-dimensional matrix algebras and structure of cones of k-positive maps
- Scientific research in fields of the Quantum Information, focused on Higher Order Transformations of Unitary Operations using Quantum Circuits frameworks, quantum teleportation via Port-Base protocol and Novel Methods of Symmetry Reduction Methods
- Create complex Python and Matlab codes to verify relations of operators on finite-dimensional algebras together with optimisation methods of Semidefinite Programming
- Domestic and international conferences with oral presentations e.g. AQIS 2023 Seoul and YITP 2023 Kyoto

Research Publications

- T. Młynik, H. Osaka, and M. Marciniak, "Characterization of k-positive maps," Communications in Mathematical Physics, vol. 406, no. 3, p. 62, 2025, doi:10.1007/s00220-025-05250-4
- D. Ebler, M. Horodecki, M. Marciniak, T. Młynik, M. T. Quintino, and M. Studziński, "Optimal universal quantum circuits for unitary complex conjugation," IEEE Transactions on Information Theory, vol. 69, no. 8, pp. 5069–5082, Aug. 2023, issn: 1557-9654. doi:10.1109/tit.2023.3263771

Pre-published

- F. Grosshans, M. Horodecki, M. Murao, T. Młynik, M. T. Quintino, M. Studziński and S. Yoshida, "Multicopy quantum state teleportation with application to storage and retrieval of quantum programs" arXiv:2409.10393 [quant-ph]
- M. Studziński, T. Młynik, M. Mozrzyński, M. Horodecki, "Irreducible matrix representations for the walled Brauer algebra" arXiv:2501.13067 [quant-ph]

Projects and Grants

- Sonata 16 - PhD Student Position, Symmetries and Entanglement in Quantum Circuits
- NAVA - PI, Academic Exchange "International scholarship exchange of PhD candidates and academic staff"
- UGrants start 2 - PI, On a class of k-entanglement witness

Teaching activities during PhD

- Algebra - 1st year UG
- Stochastic processes - 2nd year UG

- Functional analysis - 1st year UG
- Python with the basics of algorithms - 2nd year UG
- Advanced Theoretical Mathematics in Physics - 1st year UG

**Website** <https://ug.edu.pl> | **Field of study** Physics | **Final grade** 4+ | **Level in EQF** EQF level 8 |

**Thesis** Odwzorowania k-dodatnie w Fizyce (eng. k-positive maps in physics)

30/09/2015 – 20/10/2019 Gdańsk, Poland

**MSC** University of Gdańsk

- Conducted scientific research in fields of Theory of Positive maps mainly the problem of characterisation of positive maps on low-dimensional matrix algebras and structure of cones of k-positive maps
- Create complex Python and Matlab codes to verify relations of operators on finite-dimensional algebras together together with optimisation methods of Semidefinite Programming

**Website** <https://ug.edu.pl> | **Field of study** Physics | **Final grade** 4+ | **Level in EQF** EQF level 7 |

**Thesis** Nierozkładalne odwzorowania k-dodatnie na algebrach macierzowych (eng. k- positive indecomposable maps on matrix algebras).

30/09/2012 – 20/10/2015 Gdańsk, Poland

**BSC** University of Gdańsk

- Conducted exploratory data analysis to identify patterns, trends, and insights using statistical methods and data visualisation techniques to support Flat Curve of Rotation of Our Galaxy,
- Developed theoretical insight for Monte Carlo simulation, Created interactive dashboards using Excel to visualise key performance,
- Conducted data quality assessments and consistency of Satellite data of Stars and other interstellar objects with filtering of irrelevant data

Research Publications

- P. Gnaciński and T. Młynik, "Keplerian rotation of our galaxy?" Publications of the Astronomical Society of the Pacific, vol. 129, no. 974, p. 044 101, Feb. 2017, doi: 10.1088/1538-3873/aa5c9b

**Website** <https://ug.edu.pl> | **Field of study** Physics | **Final grade** 4+ | **Level in EQF** EQF level 6 |

**Thesis** Badanie krzywej rotacji galaktyki (eng. Study of galaxy's rotation curve)

02/09/2008 – 20/06/2012 Gdańsk, Poland

**TECHNICIAN** PSB Gdańsk

- Use technical design, cost estimate and implementation documentation for subsequent phases of the construction process, maintain construction documentation in accordance with applicable regulations, organise and coordinate the execution of construction, installation and network works
- Prepare sanitary installations and municipal networks for tightness tests and technical acceptance, Perform inventory measurements, acceptances and measurements of installation works and networks

**Website** <https://psb.edu.gdansk.pl/pl> | **Field of study** Building and civil engineering | **Final grade** Cum laude | **Level in EQF** EQF level 4

## LANGUAGE SKILLS

Mother tongue(s): **POLISH**

Other language(s):

|                | UNDERSTANDING |         | SPEAKING          |                    | WRITING |
|----------------|---------------|---------|-------------------|--------------------|---------|
|                | Listening     | Reading | Spoken production | Spoken interaction |         |
| <b>ENGLISH</b> | C1            | C1      | C1                | C1                 | C1      |

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

## SKILLS

work with virtual learning environments | Microsoft Office | LaTeX (very good) | programming: Python, MATLAB and SQL | Python, Scikit-Learn, Numpy, Matplotlib | Experience with math software SageMath | MATLAB | Selenium (Web Testing) - beginner | Python (computer programming) | Data Quality, Data Validation, Data Manipulation | analyse big data | MS Office, Excel programming (VBA) | document analysis results | use office systems | Version Control Systems: Github | Proficient in Optimization Solvers: CVX, Yalmip | Calculus, linear algebra, geometry, and other fundamental concepts of mathematics | mathematical physics | Python for data analysis (NumPy, Pandas, ...) and application development (PyQt) | JIRA - Intermediate level | Teaching skills | algebra | Searching via Internet ( for any kind of Research and especially scientific Research )

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| ● <b>HOBBIES AND INTERESTS</b>                                                                                                                                                                                                                                                                                                                                           |                         |
| <b>Math in Art</b>                                                                                                                                                                                                                                                                                                                                                       |                         |
| Using mathematical equations and visualisation tools like Python Matplotlib or Blender                                                                                                                                                                                                                                                                                   |                         |
| <b>Links</b> <a href="https://www.youtube.com/watch?v=8G0R3tfj1RE&amp;list=RD8G0R3tfj1RE&amp;start_radio=1">https://www.youtube.com/watch?v=8G0R3tfj1RE&amp;list=RD8G0R3tfj1RE&amp;start_radio=1</a>   <a href="https://www.linkedin.com/feed/update/urn:li:activity:7346127945561759745/">https://www.linkedin.com/feed/update/urn:li:activity:7346127945561759745/</a> |                         |
| <b>Travel</b>                                                                                                                                                                                                                                                                                                                                                            |                         |
| <b>Music (Synthwave)</b>                                                                                                                                                                                                                                                                                                                                                 |                         |
| <b>Aviation</b>                                                                                                                                                                                                                                                                                                                                                          |                         |
| ● <b>DRIVING LICENCE</b>                                                                                                                                                                                                                                                                                                                                                 |                         |
| <b>Driving Licence: B</b>                                                                                                                                                                                                                                                                                                                                                | 30/12/2015 – 30/12/2035 |